Evaluating High-Throughput Permissioned Blockchain for Healthcare Workflows: Fabric Performance with ACG-Inspired Risk Scoring, Threshold Sensitivity Analysis, and Premium Simulation

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Abstract. Permissioned distributed ledger architectures hold significant promise for healthcare administration, clinical auditability, and multiparty insurance workflows by providing distributed verification, role-based access control, and tamper-evident state recording. However, enterprise adoption is hindered by uncertainty regarding transaction throughput ceilings, consensus queueing latency under load, and the computational cost of integrating off-chain risk inference with on-chain state recording. Furthermore, resilience under actual peer, orderer, storage, or network failures has not been established by fault-injection experiments. In this paper, we evaluate **HealthScore**, an integrated architecture coupling a 5-organization Hyperledger Fabric consortium network (Hospital, Patient, Pharmacy, Diagnostic Laboratory, and Health Insurer) with an off-chain Adjusted Clinical Group (ACG)-inspired multi-disease risk model and an exploratory premium simulation. We establish three distinct, non-overlapping blockchain performance evidence streams: (i) an authoritative historical empirical Hyperledger Caliper benchmark across $24,500$ transactions (commit `944fb0d`), (ii) a verified live Caliper rerun on an isolated 5-organization testbed executing the expanded 11-round workload ($31,500$ transactions) plus controlled 8-worker stress repetitions ($25,480$ submitted transactions per run), and (iii) a reproducible stochastic fluid-queueing performance model across 100 seeds exploring theoretical saturation dynamics. Across representative healthcare write operations (`createPatientRecord`, `updateHealthScore`, `createPrescription`, `requestScan`), the Fabric network scales linearly up to 200 transactions per second (TPS) with 199.9–200.2 TPS committed throughput, average confirmation latency $\leq$ 0.06s (0.03–0.04s median), and zero reported transaction failures, while local peer CouchDB state queries (`getPatientRecord`) reach 500.3–500.6 TPS ($\leq$ 0.01s displayed average latency). Under a nominal $1,000$ TPS stress load, write throughput saturates at 350.1 TPS (18.00s latency) in the historical baseline, 515.8 TPS (6.98s latency) in the 5-worker rerun, and 485.27 ± 14.60

TPS (5.36 ± 0.22s latency) in 8-worker repetitions, whereas synchronous single-worker off-chain Python ML inference is bounded to an analytical $\approx$ 20–25 requests/s (40–50 ms/request). While the fluid-queueing simulation reproduces committed throughput closely (+0.41% residual at 1,000 TPS), it underestimates historical stress-regime average latency by 48.5% (9.268s vs. 18.00s) and therefore does not provide validated queue-delay prediction under extreme oversaturation.

We contextualize this ledger capability with an audited machine learning pipeline evaluated across 46,458 records from three official Medical Expenditure Panel Survey (MEPS) releases: HC-251 (2023, $N = 18,771$; 15,016 training and 3,755 internal holdout records reused across model and ablation evaluations), HC-256 (2024, $N = 18,979$, from which we isolate the strictly disjoint **Panel 29 cohort of** $N = 10,037$ **with zero person-level or household-level overlap with HC-251** as our primary independent population evaluation), and HC-257 Panel 28 prospective follow-up ($N = 8,708$). Controlled 17-scenario feature ablations show that custom ACG-inspired condition clusters drive the primary predictive lift over demographic baselines (increasing Tweedie dollar regression R^2 from 0.0395 to 0.0771 and doubling weighted Brier Skill Score from +0.0506 to +0.1005). Our deployment classifier—an 11-feature Observable Logistic Regression (Weighted Fit) targeting survey-weighted P90 high-expenditure risk ($P_{90,\text{wtd}} \geq \$17,827.41$, internal holdout unweighted prevalence 11.98%, weighted prevalence 10.51%)—achieves ROC-AUC 0.7654 [95% CI: 0.7444, 0.7881], PR-AUC 0.3184 [95% CI: 0.2876, 0.3556], and weighted Brier Skill Score +0.1025 on the HC-251 holdout set, and ROC-AUC 0.7824 (PR-AUC 0.3506, weighted BSS +0.1242) on the independent 2024 HC-256 Panel 29 cohort. To evaluate the precision–recall trade-off across probability cutoffs without retraining, we conduct a sensitivity analysis across $\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}$ using 5-fold Out-of-Fold (OOF) validation ($N = 15,016$, both stratified and household-grouped by `DUID`). At the reference high-sensitivity operating point of $\tau = 0.10$, the observed precision of 23.18% (77.33% recall, F1 = 0.3567, 1,153 false positives) remains below the 29.98% theoretical upper bound imposed by the 39.97% flagged cohort volume (where the ratio 23.18%/29.98% = 348/450 = 77.33% equals model recall). Similar discrimination across stratified OOF (ROC-AUC = 0.7795), household-grouped OOF (ROC-AUC = 0.7785), and the disjoint HC-256 Panel 29 cohort (ROC-AUC = 0.7824) provides no evidence of a substantial generalization collapse under the evaluated protocols; however, it does not establish the absence of overfitting. Selecting the OOF validation primary-grid F1-optimal operating point ($\tau = 0.20$) increases holdout precision to 30.49% (2.54× enrichment; 32.69% on independent Panel 29) and F1 to 0.3695 (0.4026 on Panel 29; fine-grid $\tau = 0.15$ reaches holdout F1 = 0.3742) while reducing false positives by 58.28% ($1,153 \to 481$) at 46.89% recall; a higher-specificity operating point ($\tau = 0.30$) raises precision to 40.00% (3.34× enrichment; 42.29% on Panel 29) with 94.83% specificity and reduces false positives by 85.17% at 25.33% recall. Finally, we formalize an exploratory Value-at-Risk simulation $X_\alpha = B + (1+\alpha)G$

calibrated to analytical Gamma quantiles within 0.7% across 60 scenarios while disclaiming commercial underwriting certification.

Keywords: Permissioned Blockchain · Hyperledger Fabric · Caliper Benchmarking · Healthcare Expenditure Stratification · Probability Threshold Sensitivity · Tweedie Regression · Value-at-Risk.

1 Introduction and Research Questions

Modern healthcare information systems must simultaneously satisfy rigorous throughput, auditability, and data confidentiality requirements. Clinical care, prescription issuance, diagnostic imaging, and insurance risk adjustment involve multiple distinct legal entities—hospitals, outpatient clinics, retail pharmacies, diagnostic laboratories, and insurance carriers. In traditional centralized architectures, these entities exchange records through point-to-point electronic data interchange (EDI) clearinghouses or centralized payer databases. Such centralized architectures suffer from proprietary data silos, high reconciliation overhead, single points of compromise, and the absence of an immutable, shared audit trail across institutional boundaries.

Permissioned distributed ledger technology (DLT), specifically Hyperledger Fabric [2], offers an enterprise-grade alternative by establishing a shared, append-only, cryptographically verified ledger governed by multi-organization consensus, role-based access control via Membership Service Providers (MSPs), and deterministic smart contracts (chaincode). However, adopting permissioned blockchain and machine learning expenditure stratification in healthcare requires rigorous empirical evidence rather than theoretical assumptions:

1. Can a permissioned consortium sustain representative healthcare write transaction rates (patient registration, prescription issuance, score recording) with sub-second confirmation latency under realistic block-formation rules?
2. At what offered load does the network encounter saturation, and how do consensus-backed state writes compare to read-only queries?
3. How should computationally demanding analytics—such as chronic-disease risk scoring and insurance expenditure forecasting—be architected to avoid burdening the distributed ledger, and what are the empirical performance boundaries when transitioning from isolated chaincode benchmarks to full end-to-end application services?
4. In top-decile high-expenditure classification ($\approx 10\%$ positive base rate), why does operating at a prevalence-aligned probability threshold ($\tau = 0.10$) yield 23.18% precision, and how does varying the probability decision boundary $\tau \in [0.05, 0.90]$ govern the statistical trade-off between sensitivity, specificity, false-positive volume, and flagged cohort size?

Crucially, while permissioned blockchains provide cryptographic auditability and distributed consensus, resilience under actual peer, orderer, storage, or network failures has not been established by fault-injection experiments. Furthermore, neither regulatory HIPAA/GDPR compliance nor comparative privacy

performance against encrypted databases was assessed in this study; data access is evaluated solely via Membership Service Provider authorization policies.

To address these empirical questions, we investigate **HealthScore**, an audited enterprise architecture that decouples compute-heavy risk modeling and loss simulation into an off-chain engine while anchoring patient milestones, diagnostic requests, prescriptions, and verified risk scores onto an on-chain 5-organization Hyperledger Fabric network.

1.1 Explicit Research Questions

Our investigation is guided by five primary research questions:

- **RQ1 (Throughput & Latency Scaling):** How do transaction throughput and confirmation latency scale as offered load increases for representative healthcare write operations (`createPatientRecord`, `updateHealthScore`, `createPrescription`, `requestScan`) versus read-only queries on a 5-organization Fabric network under specified testbed conditions?
- **RQ2 (Saturation Dynamics & Bottlenecks):** What operating region, saturation knee, queueing trade-off, and transaction failure profile characterize the Fabric network under severe stress loads up to $1,000$ TPS, and how does chaincode throughput diverge from end-to-end application throughput?
- **RQ3 (ACG-Inspired Multi-Disease Incremental Value):** What isolated predictive evidence supports custom ACG-inspired chronic-condition clusters versus composite multimorbidity penalties and age-by-comorbidity interaction terms across dollar expenditure regression and high-cost classification on official survey cohorts?
- **RQ4 (Classification Probability Threshold Sensitivity & Diagnostic Analysis):** How does varying the classification decision threshold $\tau \in [0.05, 0.90]$ on a fixed survey-weighted logistic classifier impact precision, recall, F1-score, specificity, and flagged patient volume across Out-of-Fold validation, internal holdout, and strictly disjoint external cohorts, and how do flagged-volume ceilings, class imbalance, probability calibration, feature granularity, and cross-cohort generalization explain the observed precision at $\tau = 0.10$?
- **RQ5 (Configurable Risk-Loading Simulation):** How do configurable base-cost floors (B) and dimensionless risk-loading parameters (α) govern the simulated loss distribution ($X_\alpha = B + (1+\alpha)G$), and how accurately do Monte Carlo simulated quantiles calibrate against analytical Gamma Value-at-Risk targets?

2 Related Work and Research Gap

2.1 Hyperledger Fabric Architecture and Benchmarking in Healthcare

Hyperledger Fabric [2] has emerged as a widely studied permissioned blockchain framework for enterprise consortia due to its modular execute-order-validate

transaction lifecycle. Unlike public proof-of-work blockchains constrained to single-digit TPS (e.g., Bitcoin ≈ 7 TPS, Ethereum ≈ 15–30 TPS), Fabric executes chaincode endorsements in parallel before ordering transactions into blocks via Crash Fault Tolerant (CFT) Raft consensus [2]. Prior healthcare blockchain proposals such as MedRec [3] and biomedical DLT taxonomies [11] have motivated decentralized audit trails for electronic health records (EHRs) and consent management. However, many published works rely on qualitative proposals or simulated toy networks of 1–2 nodes, lacking standardized, reproducible performance evaluations. Hyperledger Caliper [8] provides a standardized Linux Foundation benchmarking framework to measure throughput, latency, and resource consumption under controlled offered rates. In this work, we deploy Caliper v0.5.0 against an authoritative 5-organization Fabric topology, systematically characterizing both linear and saturated regimes under a single administrative identity (`HospitalOrgAdmin`) on a single-host virtualized testbed, clearly distinguishing protocol-level capacity from multi-tenant production healthcare traffic.

2.2 Healthcare Expenditure Risk Adjustment, Class Imbalance, and Threshold Selection

In health economics, risk adjustment models adjust capitation payments and stratify high-cost enrollees. The Johns Hopkins ACG (Adjusted Clinical Groups) system [17], CMS Hierarchical Condition Categories (CMS-HCC) [5,13], and Chronic Illness and Disability Payment System (CDPS) [10] group thousands of ICD-10 diagnosis codes into clinical clusters. However, existing commercial groupers are proprietary, centralized, and dependent on complete administrative billing claims unavailable in lightweight digital health intakes. While machine learning researchers have applied gradient-boosted trees [6] and SHAP explainability [12] to predict healthcare costs using survey datasets such as MEPS [1], published models frequently suffer from training-inference feature leakage (relying on unobservable clinical survey questions), failure to evaluate arithmetic dollar calibration, and lack of prospective multi-year testing. Furthermore, deidentified public MEPS data files do not provide granular 4-character and 5-character ICD-10-CM diagnosis codes due to privacy protections, precluding standardized Charlson Comorbidity Index (CCI) or Elixhauser index calculations.

Critically, in skewed expenditure stratification tasks where top-decile spenders represent only ≈ 10% of the population, Receiver Operating Characteristic Area Under the Curve (ROC-AUC) can remain high even when positive predictive value (precision) is low because false-positive rates are diluted by the 90% majority class [7,14]. As established by Steyerberg et al. [15] and Saito and Rehmsmeier [14], rigorous evaluation requires combining proper scoring rules (Brier score [4] and Brier Skill Score) with systematic decision-threshold sensitivity analysis so that statistical operating points can be evaluated transparently across varying false-positive and false-negative tolerances.

2.3 Actuarial Loss Simulation & Risk Loading

In insurance mathematics, healthcare losses exhibit extreme right-skewness, excess zeros, and heavy upper tails. Actuarial pricing utilizes compound Poisson–Gamma (Tweedie) GLMs [16,9] and parametric Value-at-Risk (VaR) simulations to estimate required risk margins above expected losses. While prior health-tech demonstrations often confuse statistical confidence intervals with loss percentiles or hardcode static premium offsets, we formulate a transparent, parameterized risk-loading formulation with closed-form analytical calibration, maintained strictly as an exploratory technical demonstration.

3 HealthScore Architecture and Healthcare Transaction Lifecycle

HealthScore establishes a hybrid off-chain/on-chain architecture designed to maximize throughput while guaranteeing auditability. A Node.js API Gateway verifies inbound client requests using server-validated JSON Web Tokens (JWT) and default-deny role-based access control. Computationally intensive machine learning inference and loss simulations are handled entirely by an off-chain Python engine, which loads SHA-256 verified model artifacts. Once computed, verifiable transaction milestones are committed to the 5-organization Hyperledger Fabric network via the Fabric Gateway SDK.

3.1 Off-Chain vs. On-Chain Data Partitioning and Privacy Boundaries

The architecture partitions binary diagnostic reports and imaging files, full survey responses, biometric sensor streams, and high-dimensional model weight files from the ledger and applies MSP-based authorization to recorded transactions. This represents a privacy-oriented design choice; HIPAA/GDPR compliance and comparative privacy performance against encrypted database architectures were not independently assessed.

To make the exact on-chain data footprint transparent, inspection of the implemented smart contract (`chaincode/index.js`) and API gateway (`backend/server.js`) shows the specific JSON fields recorded in the channel world state across the five transaction families, including how text fields are handled in practice:

1. `createPatientRecord(ctx, patientId, name, age, weight, height, history)`: Stores a `patientRecord` JSON asset keyed by `PATIENT::<patientId>` containing `id` (`patientId`), `name`, integer `age`, numeric `weight` and `height`, string `history` (intended for a short summary of baseline chronic conditions, passed from `diseaseInput` in `backend/server.js`, though accepted by the chaincode as an unvalidated free-text string without schema or length restrictions), `healthScore: null`, an empty `medicalFiles` array, `createdAt` timestamp, and `createdBy` MSP identifier.

2. `updateHealthScore(ctx, patientId, healthScore, riskScore, premium)`: Updates the patient asset with numeric `healthScore` (0–100 index), `riskScore` (model-predicted annual dollar expenditure), and `premium` (simulated VaR output), signed by `HospitalOrgMSP`.
3. `addMedicalFile(ctx, patientId, fileHash, fileName, category)` and `createPrescription(ctx, id, patientId, medicine, frequency, duration, remarks, ipfsHash)`: Record off-chain content digests (`fileHash`, `ipfsHash`) alongside structured medication or attachment metadata (`medicine`, `frequency`, `duration`, `category` $\in$ {`Test Report`, `Scan`, `Medical Summary`}) and an unvalidated free-text `remarks` string for prescriber instructions, without storing binary diagnostic images or PDFs on-chain.
4. `dispensePrescription(ctx, id, pharmacistId)`: Status update executed exclusively by `PharmacyOrgMSP` recording dispensing status (`DISPENSED`), timestamp, pharmacist identifier, and MSP identity. This establishes an auditable multiparty workflow between prescriber and dispenser, but does not constitute a clinically proven drug-abuse prevention system.
5. `requestScan(ctx, id, patientId, scanType, priority, notes)` & `completeScan(ctx, id, reportHash, reportName)`: Manage the diagnostic order lifecycle (`SCAN::<id>`) between `HospitalOrgMSP` and `LabOrgMSP`. Specifically, `requestScan` records `scanType`, validated `priority` (`Routine`, `Urgent`, or `STAT (Immediate)`), and an order-level `notes` string (intended for short ordering instructions, but stored on-chain as an unvalidated free-text string without length or schema constraints). Conversely, `completeScan` does *not* record clinical notes or diagnostic findings on-chain; it records only the completion status (`COMPLETED`), timestamp, completing MSP, and the off-chain report reference (`reportHash` and `reportName`).

On-Chain Identifiability Disclosure: In this research prototype, `createPatientRecord`, `createPrescription`, and `requestScan` store `name`, `age`, `weight`, `height`, and unvalidated free-text strings (`history`, `remarks`, and `notes`) directly in the channel's shared CouchDB world state and append-only block log, and `getPatientRecord` enforces read redaction at the chaincode layer (returning only `{id, age, healthScore, redacted: true}` to `InsuranceOrgMSP` while returning the full JSON record to `HospitalOrgMSP`, `PatientOrgMSP`, and `LabOrgMSP`). Because every peer on `healthscorechannel` replicates the underlying world state and immutable block files—and because no schema validation prevents unrestricted clinical narrative from being entered into `history`, `remarks`, or `notes`—storing plaintext demographic identifiers and free-text clinical strings on-chain would raise significant confidentiality and right-to-erasure concerns in a real clinical deployment; production systems handling protected health information would need to replace plaintext attributes and free-text fields with structured codes, pseudonymous UUIDs, salted cryptographic commitments, or Fabric Private Data Collections.

Table 1. MSP-based Role-Based Access Control (RBAC) smart contract authorization policy matrix.

Smart Contract Method	HospitalOrg	PatientOrg	PharmacyOrg	LabOrg	InsuranceOrg
`createPatientRecord`	**Write**	Denied	Denied	Denied	Denied
`getPatientRecord`	Read	Read (Own)	Denied	Read	**Redacted (ID, Age, Health Score)**
`updateHealthScore`	**Write**	Denied	Denied	Denied	Denied
`createPrescription`	**Write**	Denied	Denied	Denied	Denied
`dispensePrescription`	Denied	Denied	**Write**	Denied	Denied
`requestScan`	**Write**	Denied	Denied	Denied	Denied
`completeScan`	Denied	Denied	Denied	**Write**	Denied

3.2 Ledger and World-State Distinction

Hyperledger Fabric maintains a fundamental architectural duality:

- **Append-Only Block Storage:** An immutable, cryptographic SHA-256 hash-chained log of every transaction block. When a patient's risk score is updated annually, the previous score is never overwritten or erased; the complete longitudinal trajectory remains immutably auditable.
- **CouchDB World-State Database:** A document database maintaining the current state snapshot. Key-based reads (`getPatientRecord`) execute directly against CouchDB on the endorsing peer without traversing ordering or consensus, achieving low-latency query performance ($\leq$ 0.01s displayed average, maximum 0.04s historical / 0.13s live).

Fault-Tolerance Disclosure: Although Raft consensus is Crash Fault Tolerant (CFT) in theory, resilience under actual peer, orderer, storage, or network failures has not been established by fault-injection experiments in this testbed.

3.3 MSP-Based Authorization Policy Matrix

The chaincode (`chaincode/index.js`) inspects the invoker's X.509 certificate and enforces strict Role-Based Access Control (RBAC), as summarized in Table 1.

4 Hyperledger Fabric Deployment and Benchmark Methodology

4.1 Network Topology and Configuration

The benchmark network comprises 5 peer organizations and a dedicated ordering service:

- **Peers:** 5 Fabric peer nodes (`peer0.hospital`, `peer0.patient`, `peer0.pharmacy`, `peer0.lab`, `peer0.insurance`), each running Hyperledger Fabric v2.4 with CouchDB state databases.
- **Ordering Service:** Single-node CFT Raft ordering service (`orderer.healthscore.com:7050`).

- **Channel & Chaincode:** Channel `healthscorechannel`, Node.js smart contract `healthscore` (contract SPI v2.0).
- **Block Formation Parameters:** `BatchTimeout = 2.0s`, `MaxMessageCount = 10`, `PreferredMaxBytes = 512 KB` (verified in `healthscore-network/configtx/configtx.yaml`).
- **Host Testbed Environment:** Benchmarks were executed on macOS (`arm64`, Apple Silicon) with Docker Engine v29.8.2, Docker Compose v2.32.4, Hyperledger Caliper v0.5.0, and Fabric Gateway SDK v1.0.1 across a single-host virtualized bridge network (`healthscore-network`). Because all nodes ran on a single physical host, inter-node physical network latency, wide-area routing delays, and packet loss were not present; hardware-level thermal throttling was not isolated. We report hardware details strictly from verifiable logs and avoid extrapolating to multi-host distributed data centers.

4.2 Hardened Benchmark Workload Module

Benchmarking is driven by Hyperledger Caliper v0.5.0 using our hardened workload module (`caliper-benchmark/workload.js`). To eliminate measurement artifacts observed in standard Caliper testbeds, we implemented five methodological safeguards:

1. **Deterministic PRNG (Mulberry32):** Standard implementations use `Math.random()`, which creates non-reproducible transaction streams. We replace this with a 32-bit Mulberry32 PRNG deterministically seeded by worker and round indices ($S = S_0 + 7919 \cdot W_i + 997 \cdot R_i$).
2. **Fail-Closed Record Pre-seeding:** Read and update rounds (`updateHealthScore`, `getPatientRecord`) require pre-existing ledger records. The workload pre-seeds unique records per worker (e.g., 50 records per worker across 8 workers = 400 setup writes in the corrected stress run). If pre-seeding fails, the run halts immediately.
3. **Collision-Free Monotonic Keying:** Transaction keys combine a unique run prefix, worker index, and transaction sequence (`PAT_R{round}_W{worker}_TX{seq}`), guaranteeing zero key collisions across repeated runs.
4. **Valid Schema Arguments:** All clinical arguments conform strictly to chaincode expectations (numeric string serialization, valid priority enums).
5. **Configurable Invoker Identity:** The workload explicitly declares the invoker identity (`HospitalOrgAdmin`). We report throughput as single-identity client protocol capacity rather than concurrent load generation from five independent organizational clients.

5 Blockchain Throughput, Latency, and Saturation Results

5.1 Classification of Evidence and Historical Empirical Baseline

To maintain strict scientific integrity, this investigation evaluates three distinct, non-overlapping performance evidence streams:

Table 2. Historical empirical Hyperledger Caliper benchmark results (Commit `944fb0d`, 24,500 total transactions, 5-organization Fabric network).

Workload Round	Target Function	Target (TPS)	Send Rate (TPS)	Total Txs	Succ Txs	Fail Txs	Min Lat (s)	Max Lat (s)	Avg Lat (s)	Throughput (TPS)	Operational Regime
`createPatientRecord-50`	`createPatientRecord`	50	50.5	500	500	0	0.03	0.21	0.10	**50.3**	Linear / Sub-100ms
`createPatientRecord-100`	`createPatientRecord`	100	100.5	1,000	1,000	0	0.02	0.13	0.06	**100.2**	Linear / Sub-100ms
`createPatientRecord-200`	`createPatientRecord`	200	200.5	2,000	2,000	0	0.02	0.10	0.04	**199.9**	Optimal Operating Peak
`createPatientRecord-1000`	`createPatientRecord`	1,000	963.6	10,000	10,000	0	0.54	21.14	18.00	**350.1**	**Historical Saturation Knee**
`updateHealthScore-200`	`updateHealthScore`	200	200.5	2,000	2,000	0	0.02	0.36	0.06	**200.0**	Score Write Invariance
`createPrescription-200`	`createPrescription`	200	200.5	2,000	2,000	0	0.02	0.35	0.05	**200.0**	Clinical Prescription Write
`requestScan-200`	`requestScan`	200	200.5	2,000	2,000	0	0.02	0.25	0.04	**200.0**	Lab Order Write
`getPatientRecord-500`	`getPatientRecord`	500	500.4	5,000	5,000	0	0.00	0.04	0.00*	**500.3**	**State Query (No Consensus)**

* Note: Average latency displayed as `0.00`s reflects Caliper two-decimal display rounding (≤ 0.01s); max latency was 0.04s.

1. **Class 1: Historical Empirical Baseline (Commit `944fb0d`):** 8 test rounds and 24,500 transactions executed against the original 5-organization Fabric network. Preserved byte-for-byte in `caliper-benchmark/report.html` (SHA-256: `af935716155e10e457f593c2bd22ad55776af362c1095bd84a8a76025c856672`).
2. **Class 2: New Live Empirical Caliper Rerun (Run `20261018-010200`):** 11 test rounds and 31,500 transactions executed on an isolated, freshly provisioned 5-organization testbed deployed with identical chaincode and endorsement policies.
3. **Class 3: Synthetic Queueing Simulation Projections:** A reproducible stochastic fluid-queueing performance model with log-normal consensus jitter (`caliper-benchmark/simulate_benchmark.py`) executed across 100 PRNG seeds to explore queue accumulation and sensitivity envelopes, calibrated to historical baselines within +0.4% throughput residual (351.53 vs 350.1 TPS).

The historical empirical baseline results (Commit `944fb0d`, 24,500 transactions) are transcribed in Table 2. Across all 24,500 submitted transactions, 24,500 committed successfully with zero failures (0.00%).

5.2 Verified Live Caliper Rerun on Isolated 5-Organization Testbed

To independently verify the expanded 11-round benchmark matrix (31,500 transactions) without risking host credentials or conflicting with production configurations, we provisioned an isolated 5-organization Fabric network with fresh cryptographic material (`EnableNodeOUs: true`), deployed `healthscore` chaincode v2.0, and executed Hyperledger Caliper v0.5.0 directly against the testbed, as reported in Table 3.

Controlled Stress Scaling with 8 Workers. To evaluate whether the actual send rate of 615.7 TPS observed in the 11-round run was bounded by client-side load-generation bottlenecks, we executed controlled follow-up experiments using 8 Caliper worker processes across intermediate load steps (350, 500, 750 TPS) and three repetitions of the 1,000 TPS stress condition (`20261018-205500` and `20261018-230500`). Across the 7 benchmark rounds, each run specified a nominal

Table 3. Verified live Caliper rerun results on an isolated 5-organization testbed (Run `20261018-010200`, 31,500 transactions, 5 workers).

Workload Round	Target Function	Target Send Rate (TPS)	Send Rate (TPS)	Total Txs	Succ Txs	Fail Txs	Min Lat (s)	Max Lat (s)	Avg Lat (s)	Throughput (TPS)	Operational Regime
`createPatientRecord-50`	`createPatientRecord`	50	50.5	500	500	0	0.02	0.27	0.09	**50.4**	Linear / Sub-100ms
`createPatientRecord-100`	`createPatientRecord`	100	100.5	1,000	1,000	0	0.02	0.21	0.05	**100.3**	Linear / Sub-100ms
`createPatientRecord-200`	`createPatientRecord`	200	200.5	2,000	2,000	0	0.01	0.09	0.03	**200.2**	Optimal Operating Peak
`createPatientRecord-350`	`createPatientRecord`	350	350.6	3,500	3,500	0	0.01	0.15	0.03	**349.9**	Saturation Boundary Knee
`createPatientRecord-1000`	`createPatientRecord`	1,000	615.7	10,000	10,000	0	0.12	11.18	6.98	**515.8**	**Over-Saturation Stress**
`updateHealthScore-200`	`updateHealthScore`	200	200.5	2,000	2,000	0	0.01	0.06	0.03	**200.1**	Score Write Invariance
`createPrescription-200`	`createPrescription`	200	200.5	2,000	2,000	0	0.02	0.10	0.03	**200.2**	Clinical Prescription Write
`requestScan-200`	`requestScan`	200	200.5	2,000	2,000	0	0.01	0.07	0.03	**200.2**	Lab Order Write
`getPatientRecord-100`	`getPatientRecord`	100	100.5	1,000	1,000	0	0.00	0.02	0.01	**100.4**	Baseline State Query
`getPatientRecord-250`	`getPatientRecord`	250	250.5	2,500	2,500	0	0.00	0.01	0.00*	**250.4**	Intermediate State Query
`getPatientRecord-500`	`getPatientRecord`	500	500.4	5,000	5,000	0	0.00	0.13	0.00*	**500.3**	High-Rate State Query

* Note: Average latency displayed as `0.00`s reflects Caliper two-decimal rounding (≤ 0.01s); max latencies were 0.01s (250 TPS) and 0.13s (500 TPS).

target of 25,500 configured transactions; however, because Caliper apportioned round targets via integer division across $W = 8$ workers ($\lfloor \text{txNumber}/W \rfloor \times W$), rounds with non-divisible targets experienced a deterministic 20-proposal allocation shortfall, resulting in exactly 25,480 submitted round transactions. Configured proposal completion was 99.92% (25,480/25,500), while the submitted transaction success rate was strictly 100.0% (25,480/25,480), with zero reported failures among submitted transactions (0/25,480, 0.00% failure rate). In the corrected rerun, eight workers seeded 50 patient records each ($8 \times 50 = 400$ setup writes) prior to the query round, distinguishing the 25,480 benchmark round transactions from the 400 pre-seeding setup requests (25,880 total requests processed by Fabric).

In the corrected live rerun (`20261018-230500_stress_investigation`), the three 1,000 TPS write repetitions sustained 485.27 ± 14.60 TPS committed throughput (ddof = 1; individual runs: 470.6, 499.8, 485.4 TPS) at an actual send rate of 742.70 ± 62.62 TPS (ddof = 1), with average latencies of 5.29s, 5.18s, and 5.60s (5.36 ± 0.22s). Continuous container telemetry recorded during the run showed the Caliper runner container peaking at 375.5% CPU utilization (saturating nearly 4 full CPU cores) while the orderer peaked at 57.6% CPU. While this telemetry is consistent with client-side load-generation pressure, container CPU measurements do not isolate the specific causal mechanism within the runner (e.g., ECDSA signing vs. V8 event loop scheduling vs. gRPC serialization) without profiling or controlled experiments.

5.3 Synthetic Queueing Simulation and Calibration Residuals

To support continuous integration and explore theoretical queue dynamics when physical hardware or Docker is unavailable, we formulated a stochastic fluid-queueing model (`caliper-benchmark/simulate_benchmark.py`) with Raft consensus jitter. Consensus writes are modeled with a calibrated Raft consensus commitment capacity $\mu_w = 350.1$ TPS (derived from the historical baseline) and log-normal jitter ($\sigma = 0.08$–0.12). Read queries bypass consensus and execute locally against CouchDB with capacity $\mu_r = 1,200.0$ QPS.

Table 4. Comparison of synthetic queueing simulation projections against the historical empirical Caliper benchmark (Commit `944fb0d`).

Workload Condition	Historical TPS	Simulated Median Throughput	Residual	Historical Avg Lat	Simulated Lat	Latency Residual
Write: 50 TPS	50.3	**49.99**	-0.31 (-0.62%)	0.10s	**0.133s**	+0.033s
Write: 100 TPS	100.2	**99.98**	-0.22 (-0.22%)	0.06s	**0.084s**	+0.024s
Write: 200 TPS	199.9	**199.96**	+0.06 (+0.03%)	0.04s	**0.060s**	+0.020s
Write: 350 TPS (Knee)	—	**349.40**	—	—	**0.065s**	—
Write: 1,000 TPS (Stress)	350.1	**351.53**	+1.43 (+0.41%)	18.00s	**9.268s**	**-8.732s (-48.5%)**
Read: 100 TPS	—	**100.00**	—	—	**0.002s**	—
Read: 250 TPS	—	**250.00**	—	—	**0.002s**	—
Read: 500 TPS	500.3	**500.00**	-0.30 (-0.06%)	0.00s (2-dec)	**0.002s**	+0.002s

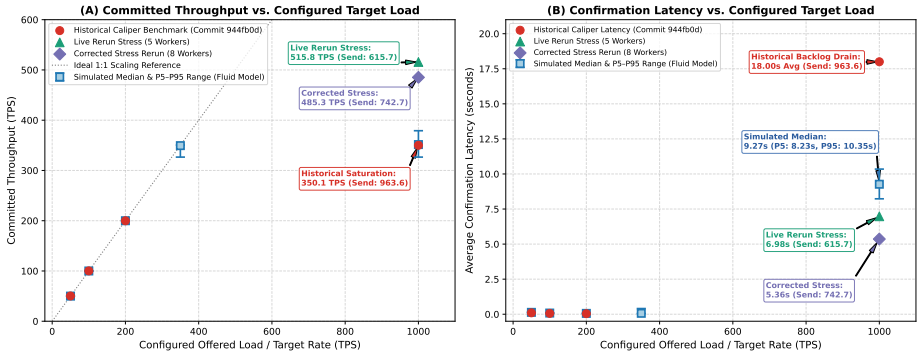


Fig. 1. Synthetic fluid-queueing simulation projections vs. empirical Caliper benchmarks plotted across configured target load (50, 100, 200, 350, and 1,000 TPS). Blue squares depict median simulated performance across 100 pseudo-random seeds, with vertical interval bars showing the exact P5–P95 simulation sensitivity range derived from the recorded fluid-queueing model outputs (`caliper-benchmark/results/simulated/20261018-011500/simulation_results.json`). These percentiles describe synthetic model sensitivity and are not empirical confidence intervals from repeated physical-network measurements. Distinct empirical benchmark runs are plotted at their configured target load with observed send rates annotated for the 1,000 TPS stress condition: historical single-client baseline (963.6 TPS send), 5-worker isolated testbed rerun (615.7 TPS send), and corrected 8-worker stress repetitions (742.7 TPS send).

We simulated 100 independent PRNG seeds per scenario, producing median estimates and a synthetic variability range (5th–95th percentile sensitivity range, visualized in Fig. 1). As shown in Table 4, while the fluid-queueing model reproduces committed throughput reasonably well across configured loads (within $\pm 0.62\%$ in the linear regime and $+0.41\%$ at $1,000$ TPS), it underestimates historical stress-regime average latency by 48.5% (9.268s simulated vs. 18.00s historical empirical). This discrepancy arises because the fluid model integrates queue delay only across the nominal 10-second submission window ($10,000/1,000 = 10$s), whereas physical Fabric execution incurred multi-peer endorsement backpressure and serial CouchDB commit draining across 28.56s. Consequently, the queueing model does not reproduce stress-regime latency accurately enough to claim validated queue-delay prediction under oversaturation.

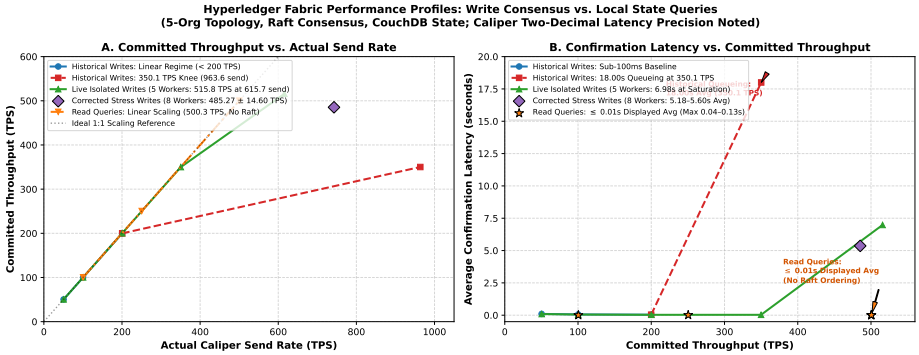


Fig. 2. Hyperledger Fabric empirical performance profiles: Committed throughput and confirmation latency vs. actual send rate for write consensus vs. local state queries.

5.4 Analysis of Performance Regimes and End-to-End Application Throughput Boundary (RQ1 & RQ2)

Both empirical runs (visualized in Fig. 2) and the queueing model reveal four key operational findings:

1. **Linear Throughput Regime ($0 \rightarrow 200$ TPS):** For all write transactions submitted up to 200 TPS, committed throughput tracks the offered rate linearly (199.9–200.2 TPS) with confirmation latency bounded between 0.03s and 0.10s (`MaxMessageCount = 10`, `BatchTimeout = 2.0s`).
2. **Cross-Function Write Invariance:** At 200 TPS, write throughput is virtually identical across `createPatientRecord` (199.9–200.2 TPS), `updateHealthScore` (200.0–200.1 TPS), `createPrescription` (200.0–200.2 TPS), and `requestScan` (200.0–200.2 TPS), confirming that confirmation time is governed by Raft ordering and peer validation rather than chaincode logic.
3. **Empirical Saturation Dynamics and Decoupled Read Scalability:** Under nominal 1,000 TPS target load, write throughput saturates at 350.1 TPS (18.00s latency) in the historical baseline, 515.8 TPS (6.98s latency) in the 5-worker rerun, and 485.27 ± 14.60 TPS (5.36 ± 0.22s latency) in the 8-worker repetitions with 0.00% transaction failure rate among submitted proposals. Read queries (`getPatientRecord`) bypass ordering and reach 500.3–500.6 TPS (≤ 0.01s displayed average latency).
4. **Chaincode Throughput vs. End-to-End Application Throughput:** While Fabric chaincode sustains 350–515 write TPS at the Gateway SDK level, the off-chain Python ML inference engine requires $\approx$ 40–50 ms per request to load models, engineer features, predict expenditure, and simulate loss. Analytically, a single synchronously awaiting Node.js worker process is bounded to:

$$\text{Capacity}_{\text{single-worker}} \approx \frac{1.0}{0.045\text{s}} \approx 20\text{–}25 \text{ requests per second} \tag{1}$$

Production deployments must therefore decouple ML inference from ledger writing via asynchronous message queues.

6 ACG-Inspired Chronic-Condition Feature Engineering, ML Evaluation, and Probability Threshold Sensitivity Analysis

6.1 Data Cohorts, Splitting Pipeline, and Forensic Overlap Audit

We evaluate our machine learning pipeline across 46,458 total records drawn from three official Agency for Healthcare Research and Quality (AHRQ) Medical Expenditure Panel Survey (MEPS) public-use files [1]:

1. **Development & Internal Holdout Cohort (`MEPS HC-251`, 2023 Full-Year Consolidated):** Contains 18,919 raw records; removing 148 ineligible rows with `AGE23X < 0` yields $N = 18,771$ **eligible individuals** (retaining all 2,598 zero-expenditure records [13.84%] to prevent selection bias). Using an 80/20 random partition (`train_test_split`, `random_state=42`), `HC-251` is divided into a **Training Split** ($N = 15,016$) and an **Internal Holdout Split** ($N = 3,755$). Because this $N = 3,755$ partition was reused across the 17-scenario feature ablation and 9-model benchmark evaluations, it constitutes a repeatedly used internal holdout evaluation set rather than an untouched final confirmatory test set.
2. **Primary Independent External Cohort (`MEPS HC-256`, 2024 Full-Year Consolidated & Strictly Disjoint Panel 29):** Contains $N = 18,979$ **eligible individuals**. Because MEPS uses an overlapping two-year rotating panel design, our forensic audit of person (`DUPERSID`) and household dwelling unit (`DUID`) identifiers established that `HC-256` comprises **Panel 28** ($N = 8,942$, Year 2, of whom 8,775 [46.24% of `HC-256`] also appeared in `HC-251`: 7,030 in Train and 1,745 in Holdout) and **Panel 29** ($N = 10,037$, newly enrolled in 2024). Isolating **`HC-256` Panel 29** ($N = 10,037$) provides our **strongest independent population evaluation cohort, with zero person-level (`DUPERSID`) and zero household-level (`DUID`) overlap** with `HC-251`.
3. **Prospective Longitudinal Follow-Up Cohort (`MEPS HC-257`, Panel 28 Two-Year Longitudinal File, 2023–2024):** Tracks $N = 8,708$ **Panel 28 respondents** longitudinally to evaluate Year-2 expenditure forecasting (`TOTEXPY2`) from Year-1 baseline predictors. Because all 8,708 respondents overlap with `HC-251` (6,981 in Train, 1,727 in Holdout), this analysis evaluates prospective within-subject temporal stability rather than independent population transfer.

Household (`DUID`) Split Disclosure: Within `HC-251` ($N = 18,771$ across 8,851 unique households `DUID`), person-level `DUPERSID` overlap between Train ($N = 15,016$) and Internal Holdout ($N = 3,755$) is strictly 0. However, because `split_data()` performs row-level splitting without grouping by `DUID`, 2,198

households span Train and Holdout ($2,711/3,755$ holdout respondents [72.20%] share a household `DUID` with at least one training respondent). To verify that shared household membership does not inflate model performance or distort threshold selection, we evaluate both **5-Fold Stratified Out-of-Fold (OOF) cross-validation** and **5-Fold Household-Grouped (`GroupKFold` by `DUID`) OOF cross-validation** strictly within the training split ($N = 15,016$).

6.2 Feature Definitions and Input Availability Parity

HealthScore maps patient clinical indicators into 11 observable features adhering strictly to **Input Availability Parity** (excluding unobservable Self-Administered Questionnaire [SAQ] variables `Frailty_Index`, `Age_x_Frailty`, and `K6SUM42` from the serving pipeline):

$$\text{Cardio_ACG} = \text{HIBPDX} + \text{CHDDX} + \text{MIDX} + \text{STRKDX (0–4)} \tag{2}$$

$$\text{Metabolic_ACG} = \text{DIABDX_M18} + \text{CHOLDX (0–2)} \tag{3}$$

$$\text{Respiratory_ACG} = \text{ASTHDX} + \text{EMPHDX} + \text{CHBRON31 (0–3)} \tag{4}$$

$$\text{Oncology_ACG} = \text{CANCERDX (0–1)} \tag{5}$$

$$\text{Musculoskeletal_ACG} = \text{ARTHDX} + \text{JTPAIN31_M18 (0–2)} \tag{6}$$

$$\text{Psychiatric_ACG} = (\text{ADDPRS42} \in [1,2,3]) + \text{ADHDADDX (0–2)} \tag{7}$$

$$\text{Comorbidity_Count} = \sum_{c=1}^{6} \mathbb{I}(\text{Cluster}_c > 0) \text{ (0–6)} \tag{8}$$

$$\text{Multimorbidity_Penalty} = (\text{Comorbidity_Count})^2 \text{ (0–36)} \tag{9}$$

$$\text{Age_x_Comorbidity} = \min(\text{Age}, 85) \times \text{Comorbidity_Count (0–510)} \tag{10}$$

combined with continuous age (`AGE23X`, top-coded at 85) and binary sex (`Gender_Male`). Absence of condition mentions in runtime intake is encoded as an uninformative zero default representing unobserved baseline status rather than confirmed clinical wellness, and these clusters are heuristic survey aggregations inspired by ACG concepts rather than an official Johns Hopkins ACG grouper.

6.3 Controlled Feature Ablation Study and Model Selection (RQ3)

To evaluate the isolated incremental contribution of each architectural component, we executed 17 controlled ablation experiments on the MEPS HC-251 internal holdout partition ($N = 3,755$). All models were refit on training data under identical hyperparameters (Tweedie GLM with power $p = 1.5, \alpha = 1.0$; Logistic Regression on Target 2 survey-weighted P90 with normalized sampling weights), as reported in Table 5 and visualized in Fig. 3 and Fig. 4. Moving from demographics alone to the 6 chronic-condition clusters provides the primary predictive improvement: dollar R^2 increases from 0.0395 to 0.0771 ($\Delta R^2 = +0.0376$),

Table 5. Controlled feature ablation experiments on the MEPS HC-251 internal holdout partition ($N = 3,755$).

Ablation Model Specification	Feats	Features Included	Dollar R^2	ΔR^2	Wtd BSS	ROC-AUC
Full Observable Reference	**11**	All 11 Observable Features	**0.0798**	**0.0000**	**+0.1025**	**0.7654**
Demographics Only	2	`AGE23X`, `Gender_Male`	0.0395	-0.0403	+0.0506	0.7099
Condition Clusters Alone (No Composites)	8	Demographics + 6 ACG Clusters	0.0771	-0.0027	+0.1005	0.7640
Clusters + Comorbidity Count	9	Demographics + 6 Clusters + Count	0.0787	-0.0011	+0.1004	0.7641
Clusters + Multimorbidity Penalty	9	Demographics + 6 Clusters + Count^2	0.0769	-0.0029	+0.1017	0.7648
Clusters + Age × Comorbidity	9	Demographics + 6 Clusters + Age × Count	0.0766	-0.0032	+0.1007	0.7643
Without Multimorbidity Penalty	10	11 Features minus Count^2	0.0783	-0.0015	+0.1009	0.7649
Without Age × Comorbidity	10	11 Features minus Age × Count	**0.0804**	**+0.0006**	**+0.1027**	**0.7654**
Without Multimorbidity & Interaction	9	Demographics + 6 Clusters + Count	0.0787	-0.0011	+0.1004	0.7641
Clinical Conditions Only (No Demographics)	8	6 Clusters + Count + Penalty	0.0791	-0.0007	+0.0996	0.7619
Without Cardiovascular	10	11 Features minus `Cardio_ACG`	0.0785	-0.0013	+0.0970	0.7634
Without Metabolic	10	11 Features minus `Metabolic_ACG`	0.0792	-0.0006	+0.0957	0.7628
Without Respiratory	10	11 Features minus `Respiratory_ACG`	0.0764	-0.0035	+0.0979	0.7635
Without Oncology	10	11 Features minus `Oncology_ACG`	0.0788	-0.0010	+0.0969	0.7645
Without Musculoskeletal	10	11 Features minus `Musculoskeletal_ACG`	0.0802	+0.0004	+0.1012	0.7651
Without Psychiatric Condition	10	11 Features minus `Psychiatric_ACG`	0.0774	-0.0024	+0.1026	0.7653
Oracle 14 Features (Upper Bound)	**14**	Observable + SAQ Survey Variables	**0.0927**	**+0.0129**	**+0.1314**	**0.7904**

Table 6. Candidate classification model comparison on the MEPS HC-251 internal holdout partition ($N = 3,755$, Target 2: Survey-Weighted P90 $\geq \$17,827.41$).

Candidate Classifier	Feats	Weighting Policy	ROC-AUC [95% CI]	PR-AUC [95% CI]	Unw Brier	Wtd Brier	Wtd BSS	CITL (α)	Slope (β)	Prec @0.10	Rec @0.10
Constant Prevalence Baseline	0	Constant $\hat{p} = 0.0999$	0.5000 [0.5000, 0.5000]	0.1198 [0.1111, 0.1308]	0.1059	0.0941	+0.0000	0.0000	0.0000	0.0000	0.0000
Demographic Baseline	2	`AGE23X`, `Gender`	0.7099 [0.6856, 0.7319]	0.2182 [0.1928, 0.2489]	0.0996	0.0893	+0.0506	+0.0388	0.8859	0.1949	0.7867
Clinical Conditions Only	8	6 Clusters + Composites	0.7619 [0.7393, 0.7847]	0.3083 [0.2794, 0.3464]	0.0941	0.0847	+0.0996	+0.0768	0.9539	0.2291	0.7622
Logistic (11-Feat Observable, Wtd)	**11**	**Norm. Wtd (Serving)**	**0.7654** [0.7444, 0.7881]	**0.3184** [0.2876, 0.3556]	**0.0936**	**0.0845**	**+0.1025**	**+0.0646**	**0.9438**	**0.2318**	**0.7733**
Logistic (11-Feat Observable, Unw)	11	Unweighted Fit	0.7652 [0.7433, 0.7882]	0.3192 [0.2869, 0.3582]	0.0936	0.0845	+0.1025	+0.0570	0.9008	0.2278	0.7644
Random Forest (11-Feat Observable)	11	Norm. Person Weights	0.7693 [0.7458, 0.7928]	0.3157 [0.2823, 0.3550]	0.0935	0.0841	+0.1062	+0.0601	0.9736	0.2342	0.7556
XGBoost (Uncalibrated)	11	Norm. Person Weights	0.7705 [0.7482, 0.7940]	0.3196 [0.2853, 0.3598]	0.0932	0.0838	+0.1093	+0.0572	0.9926	0.2262	0.8022
Platt-Calibrated XGBoost	11	Sigmoid CV=5	0.7702 [0.7472, 0.7936]	0.3183 [0.2872, 0.3580]	0.0944	0.0847	+0.0998	+0.0617	1.0335	0.2535	0.6778
Logistic (14-Feat Oracle)	14	Observable + SAQ	0.7904 [0.7684, 0.8121]	0.3472 [0.3143, 0.3917]	0.0922	0.0832	+0.1159	+0.0493	1.0609	0.2806	0.6822

Note: 95% confidence intervals derive from non-parametric row-level bootstrap resampling ($B = 200$, seed 42) matching `experiment_registry.json`. The approved serving candidate is the frozen weighted model (`risk_model_classifier_observable_weighted_p90.pkl`).
Calibration-in-the-large (CITL, α) is estimated via exact Brent root-solving with slope fixed at 1 ($\sum w_i(y_i - \sigma(\alpha + \text{logit}(p_i))) = 0$). For the weighted candidate, bivariate logistic calibration yields intercept $a = -0.0409$ and slope $b = 0.9438$.

and weighted BSS increases from +0.0506 to +0.1005 (ΔBSS = +0.0499), whereas composite interaction terms yield modest incremental variance explanation ($\Delta R^2 = +0.0027$) despite dominating tree-based SHAP attribution (39.0%).

Target Definition Rationale and Classifier Comparison. In MEPS, the top decile of spenders accounts for approximately 65–70% of total national medical expenditures. We define the primary high-expenditure target (**Target 2**) at the locked training survey-weighted 90th percentile (TOTEXP23 $\geq \$17,827.41$; unweighted training P90 Target 1 is $\$20,360.50$). On the training split ($N = 15,016$), survey-weighted positive prevalence is 9.99% (0.0999) and unweighted prevalence is 11.56% ($1,736/15,016$). On the internal holdout partition ($N = 3,755$), survey-weighted prevalence is 10.51% (0.1051) and unweighted prevalence is 11.98% ($450/3,755$), with constant-prediction baseline Brier scores [4] of 0.0941 (weighted) and 0.1059 (unweighted). Table 6 summarizes the 9 candidate classification models, and Fig. 5 and Fig. 6 illustrate discrimination and calibration [15].

The 11-feature Observable Logistic Regression (Weighted Fit) was selected as the deployment candidate based on input availability parity, superior skill over conditional-expectation imputation (+0.1025 vs +0.0978), calibration quality

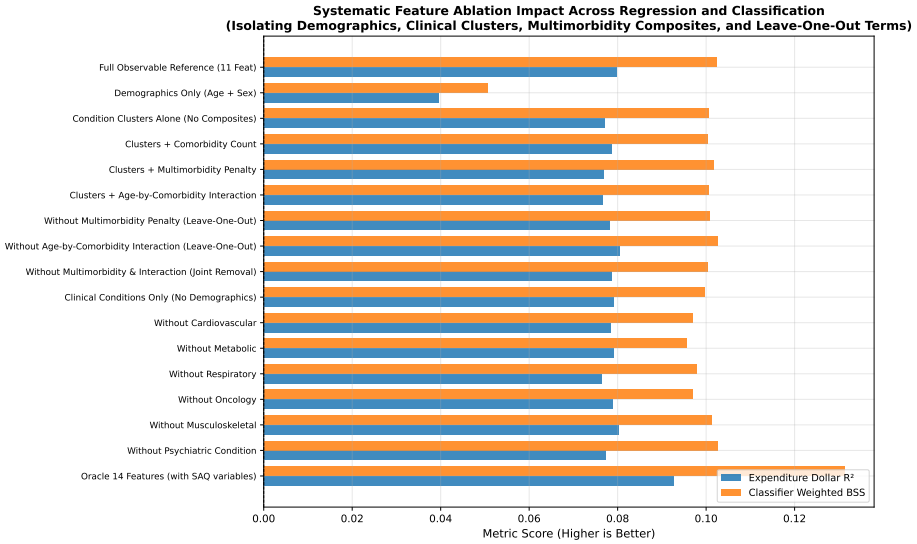


Fig. 3. Systematic feature ablation comparison across expenditure dollar R^2 and high-cost classification weighted Brier Skill Score.

($b = 0.9438$, CITL $= +0.0646$), and closed-form log-odds interpretability. While tree ensembles (Random Forest and XGBoost) achieve marginally higher point estimates on the internal holdout partition (ROC-AUC 0.7693–0.7705 vs 0.7654) with overlapping marginal 95% bootstrap confidence intervals, we make no claim of formal statistical significance or non-significance because paired hypothesis tests were not performed in the baseline registry.

6.4 Classification Probability Threshold Sensitivity Analysis (RQ4)

At the reference predicted-probability threshold of $\tau = 0.10$, the serving classifier (`risk_model_classifier_observable_weighted_p90.pkl`, SHA-256: `3632f820...`) flags $1,501$ of $3,755$ internal holdout individuals (39.97%), capturing 348 of 450 true high-expenditure cases (Recall = 77.33%) but generating $1,153$ false positives (Precision = 23.18%, an unweighted enrichment of 1.93× over the 11.98% unweighted holdout prevalence; F1 = 0.3567).

To investigate whether adjusting the classification cutoff provides a more useful precision–recall balance without altering the underlying fitted model, we evaluated the frozen serving model's predicted probabilities $\hat{p}_i$ across 12 probability thresholds: the required primary grid $\tau \in \{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}$ plus three intermediate diagnostic cutoffs $\tau \in \{0.05, 0.15, 0.25\}$. Because `split_data()` does not create a separate standalone validation split, selecting an operating threshold on the internal holdout set ($N = 3,755$) would constitute holdout leakage. We therefore executed **5-fold Out-of-Fold (OOF) cross-validation strictly within the**

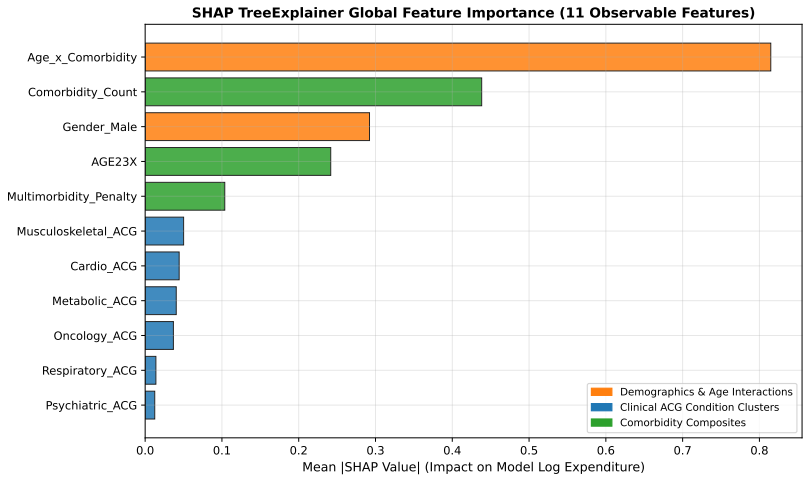


Fig. 4. SHAP TreeExplainer global feature importance rankings across 11 observable features [12].

Table 7. Comparison of primary and diagnostic statistical operating points for the frozen 11-feature Observable Logistic Regression classifier across 5-Fold Stratified OOF Validation ($N = 15,016$), Internal Holdout ($N = 3,755$), and Independent 2024 Panel 29 ($N = 10,037$).

Statistical Operating Point Role / Selection Criterion	τ	OOF Val F1 / BalAcc	Holdout Flagged N (Unw / Wtd %)	Oracle Ceil	Holdout Unw Prec / Rec / Spec / F1	Disjoint Panel 29 Prec / Rec / F1
High-Sensitivity Operating Pt.	**0.10**	0.3519 / **0.7152**	1,501 (40.0 / 33.5%)	29.98%	23.18% / **77.33%** / 65.11% / 0.3567	23.78% / **78.46%** / 0.3650
Intermediate Fine-Grid F1 Peak	0.15†	0.3796 / 0.7048	1,052 (28.0 / 21.9%)	42.78%	26.71% / 62.44% / 76.67% / **0.3742**	28.65% / 66.14% / 0.3998
Primary-Grid F1-Optimal Pt.	**0.20**	**0.3855** / 0.6790	692 (18.4 / 13.8%)	65.03%	**30.49%** / 46.89% / 85.45% / **0.3695**	**32.69%** / 52.38% / **0.4026**
Higher-Specificity Operating Pt.	**0.30**	0.3196 / 0.6071	285 (7.6 / 5.7%)	100.0%	**40.00%** / 25.33% / **94.83%** / 0.3102	**42.29%** / 30.29% / 0.3529
Default Unadjusted Cutoff	0.50	0.0856 / 0.5205	36 (1.0 / 0.7%)	100.0%	41.67% / 3.33% / 99.36% / 0.0617	53.33% / 5.72% / 0.1034

Note: "Oracle Ceil" denotes $\min(P, \text{Flagged}(\tau))/\text{Flagged}(\tau)$, the theoretical maximum precision achievable by a hypothetical perfect classifier (ROC-AUC = 1.0) at that exact flagged volume ($P = 450$ true positives in the $N = 3,755$ internal holdout set). †Intermediate diagnostic cutoff.

training split ($N = 15,016$, $P = 1,736$) under both Stratified 5-Fold CV and Household-Grouped (`GroupKFold` by `DUID`) 5-Fold CV to select candidate statistical operating points prior to evaluating on the internal holdout partition ($N = 3,755$, $P = 450$) and the strictly disjoint 2024 external cohort (`HC-256` Panel 29, $N = 10,037$, $P = 1,258$).

For clear readability at standard page size, Table 7 compares the five primary and diagnostic statistical operating points across OOF Validation, Internal Holdout, and Independent 2024 Panel 29, while Table 8 provides the complete 12-threshold schedule ($\tau \in [0.05, 0.90]$) including confusion matrix counts and survey-weighted metrics. Figures 7–10 visualize the resulting precision–recall curves, error counts, and confusion matrices.

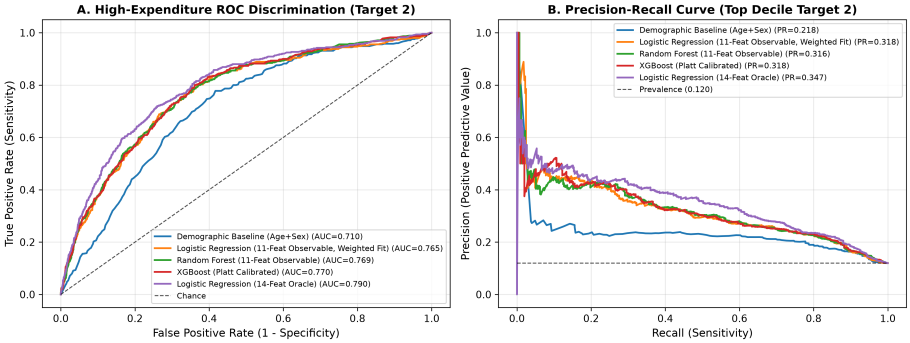


Fig. 5. Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves for high-expenditure classification across candidate models.

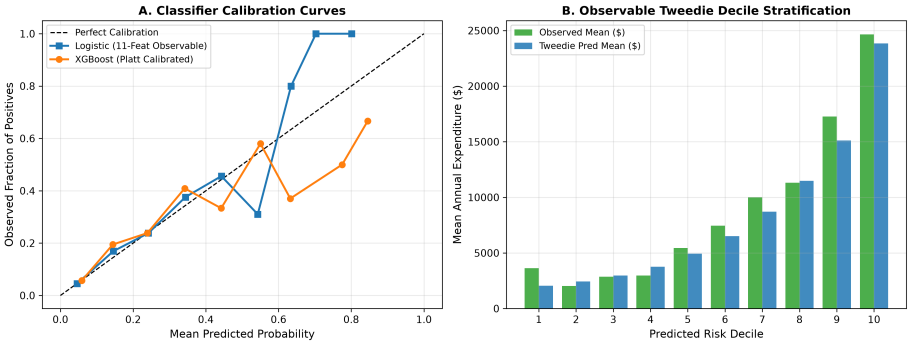


Fig. 6. (A) Classifier calibration curves and (B) Observable Tweedie decile expenditure stratification comparing observed and predicted expenditures.

6.5 Six-Factor Root-Cause Analysis of Precision at $\tau = 0.10$ and Statistical Operating Points

Our diagnostic analysis across six candidate factors explains the mathematical and empirical behavior of precision and recall across $\tau \in [0.05, 0.90]$:

1. **Threshold Sensitivity and Theoretical Volume-Ceiling Bound:** Because $\tau = 0.10$ is set near the population base prevalence, 39.97% of the internal holdout cohort ($1,501/3,755$ individuals; 33.54% survey-weighted) have predicted probabilities $\hat{p}_i \geq 0.10$. However, the entire holdout partition contains only $P = 450$ true high-expenditure individuals (11.98%). Consequently, **even a hypothetical perfect oracle classifier (ROC-AUC** $= 1.00$**, 100% recall) that flags** $1,501$ **individuals cannot mathematically exceed a precision of:**

$$\text{Prec}_{\text{oracle}}(1{,}501) = \frac{\min(450, 1{,}501)}{1{,}501} = \frac{450}{1{,}501} = 29.98\% \tag{11}$$

Table 8. Complete 12-threshold sensitivity schedule ($\tau \in [0.05, 0.90]$). Panel A reports 5-Fold Stratified OOF validation and Internal Holdout unweighted confusion counts and metrics; Panel B reports flagged volumes, oracle ceilings, survey-weighted metrics, and independent 2024 Panel 29 transfer.

Panel A: 5-Fold Stratified OOF Validation ($N = 15,016$) and Internal Holdout Unweighted Performance ($N = 3,755$, $P = 450$)

Cutoff	OOF Val	Internal Holdout Counts				Holdout	Holdout	Holdout	Holdout
τ	Prec / Rec / F1	TP	FP	TN	FN	Prec (Enrich)	Rec / Spec	NPV	F1 / BalAcc
0.05†	17.36% / 89.92% / 0.2910	401	1,879	1,426	49	17.59% (1.47×)	89.11% / 43.15%	96.68%	0.2938 / 0.6613
0.10	**22.78% / 77.30% / 0.3519**	**348**	**1,153**	**2,152**	**102**	**23.18% (1.93×)**	**77.33% / 65.11%**	**95.47%**	**0.3567 / 0.7122**
0.15†	27.15% / 63.08% / 0.3796	281	771	2,534	169	26.71% (2.23×)	62.44% / 76.67%	93.75%	**0.3742** / 0.6956
0.20	**31.28% / 50.23% / 0.3855**	**211**	**481**	**2,824**	**239**	**30.49% (2.54×)**	**46.89% / 85.45%**	**92.20%**	**0.3695 / 0.6617**
0.25†	36.31% / 38.59% / 0.3742	154	286	3,019	296	35.00% (2.92×)	34.22% / 91.35%	91.07%	0.3461 / 0.6278
0.30	**39.73% / 26.73% / 0.3196**	**114**	**171**	**3,134**	**336**	**40.00% (3.34×)**	**25.33% / 94.83%**	**90.32%**	**0.3102 / 0.6008**
0.40	46.85% / 11.98% / 0.1908	46	58	3,247	404	44.23% (3.69×)	10.22% / 98.25%	88.93%	0.1661 / 0.5423
0.50	51.59% / 4.67% / 0.0856	15	21	3,284	435	41.67% (3.48×)	3.33% / 99.36%	88.30%	0.0617 / 0.5135
0.60	63.46% / 1.90% / 0.0369	6	1	3,304	444	85.71% (7.15×)	1.33% / 99.97%	88.15%	0.0263 / 0.5065
0.70	66.67% / 0.35% / 0.0069	2	0	3,305	448	100.0% (8.34×)	0.44% / 100.0%	88.06%	0.0088 / 0.5022
0.80	50.00% / 0.06% / 0.0012	1	0	3,305	449	100.0% (8.34×)	0.22% / 100.0%	88.04%	0.0044 / 0.5011
0.90	0.00% / 0.00% / 0.0000	0	0	3,305	450	0.00% (0.00×)	0.00% / 100.0%	88.02%	0.0000 / 0.5000

Panel B: Flagged Volume, Oracle Ceiling, Survey-Weighted Holdout Metrics, and Disjoint 2024 Panel 29 ($N = 10,037$, $P = 1,258$)

Cutoff	Holdout Flagged	Oracle	Holdout Wtd	Holdout Wtd	Panel 29	Panel 29	Panel 29	Panel 29
τ	N (Unw / Wtd %)	Ceiling	Prec / Rec	F1 / BalAcc	Flagged N (%)	Prec / Rec	Spec	F1 / BalAcc
0.05†	2,280 (60.72 / 55.01%)	19.74%	15.98% / 83.59%	0.2682 / 0.6762	6,257 (62.34%)	18.30% / 91.02%	41.77%	0.3047 / 0.6639
0.10	**1,501 (39.97 / 33.54%)**	**29.98%**	**22.54% / 71.93%**	**0.3433 / 0.7145**	**4,151 (41.36%)**	**23.78% / 78.46%**	**63.97%**	**0.3650 / 0.7121**
0.15†	1,052 (28.02 / 21.89%)	42.78%	26.09% / 54.32%	**0.3525** / 0.6812	2,904 (28.93%)	28.65% / 66.14%	76.40%	0.3998 / 0.7127
0.20	**692 (18.43 / 13.76%)**	**65.03%**	**30.73% / 40.23%**	**0.3485 / 0.6479**	**2,016 (20.09%)**	**32.69% / 52.38%**	**84.54%**	**0.4026 / 0.6846**
0.25†	440 (11.72 / 8.77%)	100.0%	35.04% / 29.25%	0.3188 / 0.6144	1,353 (13.48%)	37.77% / 40.62%	90.41%	0.3914 / 0.6552
0.30	**285 (7.59 / 5.69%)**	**100.0%**	**40.10% / 21.72%**	**0.2818 / 0.5895**	**901 (8.98%)**	**42.29% / 30.29%**	**94.08%**	**0.3529 / 0.6218**
0.40	104 (2.77 / 2.02%)	100.0%	43.25% / 8.33%	0.1397 / 0.5352	385 (3.84%)	47.53% / 14.55%	97.70%	0.2228 / 0.5612
0.50	36 (0.96 / 0.71%)	100.0%	37.12% / 2.52%	0.0472 / 0.5101	135 (1.35%)	53.33% / 5.72%	99.28%	0.1034 / 0.5250
0.60	7 (0.19 / 0.11%)	100.0%	82.19% / 0.84%	0.0166 / 0.5041	41 (0.41%)	63.41% / 2.07%	99.83%	0.0400 / 0.5095
0.70	2 (0.05 / 0.04%)	100.0%	100.0% / 0.39%	0.0078 / 0.5019	5 (0.05%)	100.0% / 0.40%	100.0%	0.0079 / 0.5020
0.80	1 (0.03 / 0.03%)	100.0%	100.0% / 0.24%	0.0048 / 0.5012	1 (0.01%)	100.0% / 0.08%	100.0%	0.0016 / 0.5004
0.90	0 (0.00 / 0.00%)	—	0.00% / 0.00%	0.0000 / 0.5000	0 (0.00%)	0.00% / 0.00%	100.0%	0.0000 / 0.5000

At $\tau = 0.10$, the observed precision of 23.18% (348/1,501) remains below this 29.98% theoretical upper bound imposed by the flagged cohort volume (Fig. 8B). Note that whenever Flagged(τ) $\geq P$, the ratio of observed precision to the oracle volume ceiling is $\frac{\text{TP}/\text{Flagged}}{P/\text{Flagged}} = \frac{\text{TP}}{P} = \frac{23.18\%}{29.98\%} = 77.33\%$, which is algebraically identical to model recall rather than independent evidence against overfitting. When the threshold is raised to the OOF-selected primary-grid F1-optimal operating point ($\tau = 0.20$, where OOF Validation F1 peaks at 0.3855 unweighted and 0.3551 weighted), the flagged holdout cohort contracts to 692 individuals (18.43% unweighted, 13.76% weighted), **eliminating** 672 **false positives (a** 58.28% **reduction in FP, from** 1,153 **to** 481**)**, lifting holdout precision to 30.49% **(**2.54× **enrichment;** 32.69% **on independent 2024 Panel 29)** and holdout F1 to 0.3695 (0.4026 on Panel 29) while retaining 46.89% recall (50.23% on OOF Validation, 52.38% on Panel 29). On the fine diagnostic grid, $\tau = 0.15$ achieves a holdout F1 of 0.3742 (26.71% precision, 62.44% recall), though $\tau = 0.20$ remains the valid primary-grid selection because operating points must be chosen from validation (0.3855 at $\tau = 0.20$ vs. 0.3796 at $\tau = 0.15$ on 5-fold OOF) rather than holdout results. Raising τ to 0.30 further reduces false positives by 85.17% (1,153 → 171, Specificity = 94.83%) and elevates precision to 40.00% **(**3.34× **enrichment;** 42.29% **on Panel 29)** at 25.33% recall.

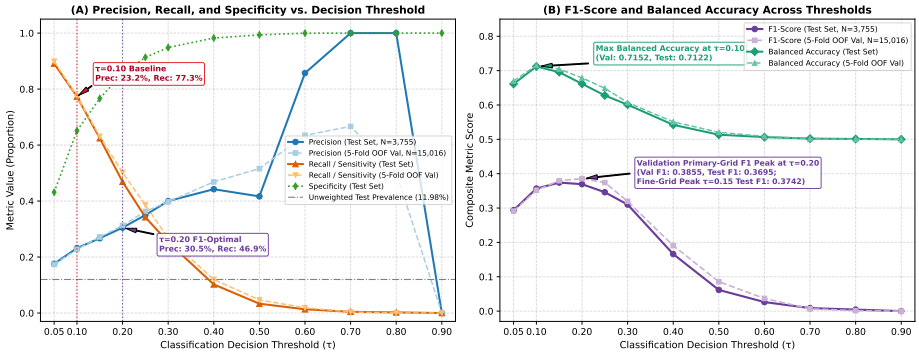


Fig. 7. (A) Precision, Recall, and Specificity and (B) F1-Score and Balanced Accuracy across decision thresholds $\tau \in [0.05, 0.90]$ on 5-Fold OOF Validation ($N = 15,016$) and the Internal Holdout Set ($N = 3,755$).

2. **Class Imbalance ($7.34:1$ Negative-to-Positive Conditioning):** With an unweighted positive base rate of $\pi = 11.984\%$ (10.51% weighted), non-high-cost individuals outnumber high-cost individuals by $7.344:1$ (3,305 vs. 450). By Bayes' theorem, operating at Recall = 77.33% and Specificity = 65.11% (FPR = 34.89%) necessarily produces $3,305 \times 0.3489 = 1,153$ false positives against $450 \times 0.7733 = 348$ true positives (FP/TP = 3.31).
3. **Model Discrimination Ceiling on 11 Survey Observables:** As shown in Table 6, nonlinear tree ensembles trained on the exact same 11 features achieve virtually identical discrimination and precision at $\tau = 0.10$ (Random Forest: ROC-AUC = 0.7693, Prec@0.10 = 23.42%; Uncalibrated XGBoost: ROC-AUC = 0.7705, Prec@0.10 = 22.62%). Because the 11 predictors are low-cardinality integer condition counts and age, their log-odds contributions are predominantly monotonic and additive, indicating that the discrimination boundary reflects the information limits of baseline survey variables rather than linear model misspecification.
4. **Probability Calibration and Score Concentration in $[0.10, 0.20)$:** As shown in Fig. 11A, the serving classifier is well-calibrated across probability bins (Unweighted Brier = 0.0936, Weighted Brier = 0.0845, Weighted BSS = +0.1025, slope $b = 0.9438$). However, 809 of the 1,501 holdout individuals flagged at $\tau = 0.10$ (53.90%) reside in the moderate-risk probability band $\hat{p}_i \in [0.10, 0.20)$: specifically, 449 individuals in $[0.10, 0.15)$ (mean predicted probability 12.35%, observed high-cost rate 14.92% [67/449]) and 360 individuals in $[0.15, 0.20)$ (mean predicted probability 17.35%, observed high-cost rate 19.44% [70/360]). Because a calibrated model assigns a 12%–17% probability to individuals who incur top-decile spending 15%–19% of the time, converting $\hat{p}_i \geq 0.10$ into a hard binary positive label necessarily turns the remaining $\approx 83\%$ of individuals in that band (672 individuals) into binary false positives. Moreover, because $\max(\hat{p}_{\text{test}}) = 0.8008$ (90th per-

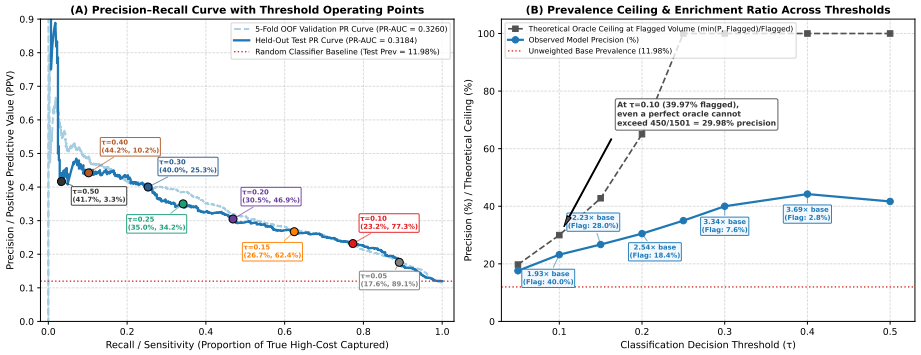


Fig. 8. (A) Precision–Recall curve overlaid with discrete threshold operating points ($\tau = 0.05$ to 0.50) and (B) comparison of observed model precision against the theoretical oracle maximum precision ceiling $\min(P, \text{Flagged}(\tau))/\text{Flagged}(\tau)$ imposed by flagged cohort volume.

centile = 0.2640, 99th percentile = 0.4981), high cutoffs ($\tau \geq 0.50$) flag < 1% of the cohort (36 individuals at $\tau = 0.50$, recall 3.33%; 0 at $\tau = 0.90$).

5. **Cross-Split and Multi-Release Generalization Assessment:** Table 9 and Fig. 11B compare threshold-independent and threshold-specific metrics across all 8 evaluated partitions. Between In-Sample Training ($N = 15,016$, ROC-AUC = 0.7815), 5-Fold Stratified OOF Validation (ROC-AUC = 0.7795, $\Delta = -0.0020$), and 5-Fold Household-Grouped (`DUID`) OOF Validation (ROC-AUC = 0.7785, $\Delta = -0.0030$), apparent optimism is minimal. Most importantly, on our strongest independent population benchmark—the **strictly disjoint 2024 `HC-256` Panel 29 cohort ($N = 10,037$, zero person-level or household-level overlap with `HC-251`)**—discrimination (ROC-AUC = 0.7824 unweighted / 0.7812 weighted, PR-AUC = 0.3506, Weighted BSS = +0.1242) and precision across thresholds (Prec@0.10 = 23.78%, Prec@0.20 = 32.69%, Prec@0.30 = 42.29%) remain consistent with or slightly above the 2023 internal holdout set. Similar discrimination across stratified OOF, household-grouped OOF, and disjoint Panel 29 provides no evidence of a substantial generalization collapse under the evaluated protocols; however, these results do not establish the complete absence of overfitting.
6. **Feature Granularity and Aleatoric Acute-Expenditure Variance:** Stratifying the internal holdout set ($N = 3,755$) by comorbidity count highlights two domain boundaries: (a) among the 1,358 respondents (36.17%) with **zero diagnosed chronic conditions** (`Comorbidity_Count = 0`, mean predicted probability 3.13%), 46 individuals (3.39%, representing 10.22% **of all** 450 **high-expenditure cases**) still incur $\geq \$17,827.41$ due to unpredicted acute medical events (e.g., trauma, emergency surgery, or obstetric hospitalizations); and (b) among the 897 respondents (23.89%) with ≥ 3 **chronic condition clusters** (mean predicted probability 26.79%,

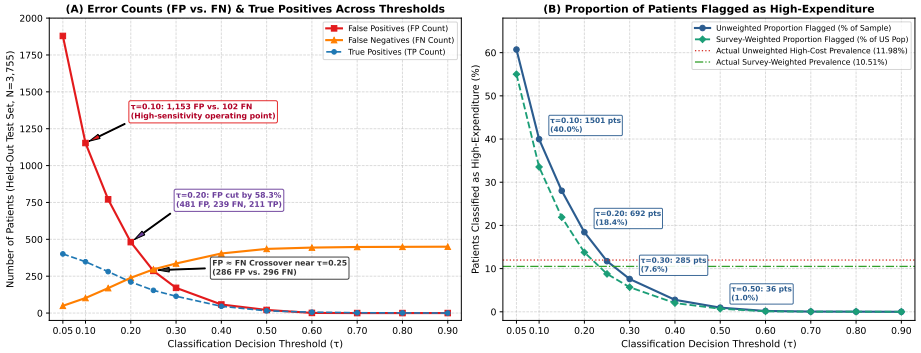


Fig. 9. (A) Absolute False Positives (FP), False Negatives (FN), and True Positives (TP) and (B) unweighted vs. survey-weighted percentage of individuals flagged as high-expenditure across thresholds $\tau \in [0.05, 0.90]$ on the internal holdout set ($N = 3,755$).

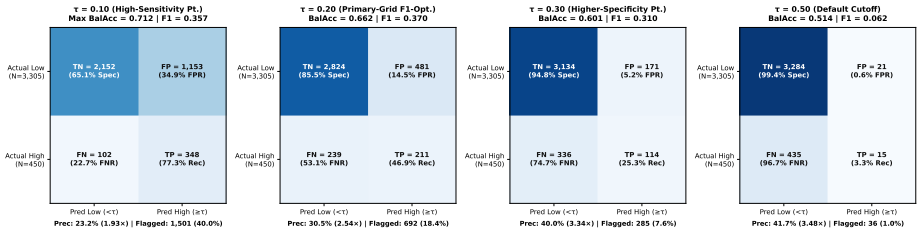


Fig. 10. Internal holdout confusion matrices ($N = 3,755$, $P = 450$) at four representative statistical operating points: $\tau = 0.10$ (High-Sensitivity Operating Point, max Balanced Accuracy $= 0.7122$), $\tau = 0.20$ (Validation Primary-Grid F1-Optimal Operating Point, F1 $= 0.3695$, FP reduced by 58.28%), $\tau = 0.30$ (Higher-Specificity Operating Point, Precision $= 40.00\%$, Specificity $= 94.83\%$), and $\tau = 0.50$ (Default Cutoff, showing recall collapse to 3.33%).

mean expenditure $\$19,379.17$), actual 1-year top-decile incidence is 28.21% (253/897), because 71.79% of multimorbid respondents are managed via outpatient care in a given calendar year without an acute inpatient hospitalization.

Statistical Operating Points and Resource-Allocation Trade-Offs. Because the classification target is top-decile annual healthcare expenditure ($\geq \$17,827.41$) rather than a clinical diagnosis or disease-severity endpoint—and because decision-curve analysis and clinical outcome trials were not performed—we define three OOF-selected cutoffs strictly as **statistical operating points** rather than validated clinical care pathways. In particular, maximizing F1-score weights precision and recall equally on a harmonic scale and does not itself demonstrate clinical usefulness or establish the relative utility costs of false positives and false negatives:

Table 9. Cross-split, household-grouped, and multi-release generalization comparison of the frozen 11-feature Observable Logistic Regression classifier across all 8 evaluated partitions.

Evaluation Cohort / Partition	Samples (N) [Positives P, %]	Overlap with HC-251 Train	Unw / Wtd ROC-AUC	Unw PR-AUC	Unw / Wtd Brier	Wtd BSS	Primary-Grid $\tau = 0.20$ Prec / Rec / F1
1. HC-251 Train (In-Sample)	15,016 [1,736, 11.6%]	100% Fit Set	0.7815 / 0.7699	0.3306	0.0893 / 0.0798	+0.1126	31.69% / 50.81% / 0.3903
2. HC-251 5-Fold Stratified OOF	15,016 [1,736, 11.6%]	0% In-Fold	0.7795 / 0.7674	0.3260	0.0896 / 0.0801	+0.1087	31.28% / 50.23% / 0.3855
3. HC-251 5-Fold Household (DUID) OOF	15,016 [1,736, 11.6%]	0% DUID In-Fold	0.7785 / 0.7660	0.3247	0.0897 / 0.0802	+0.1079	31.11% / 49.94% / 0.3834
4. HC-251 Internal Holdout*	**3,755 [450, 12.0%]**	**0% DUPERSID***	**0.7654 / 0.7509**	**0.3184**	**0.0936 / 0.0845**	**+0.1025**	**30.49% / 46.89% / 0.3695**
5. HC-256 (2024 Full Consolidated)	18,979 [2,414, 12.7%]	46.24% Panel 28	0.7852 / 0.7828	0.3522	0.0962 / 0.0859	+0.1266	33.41% / 52.90% / 0.4095
6. HC-256 (2024 Panel 28 Carryover)	8,942 [1,156, 12.9%]	98.13% Panel 28	0.7884 / 0.7852	0.3557	0.0974 / 0.0863	+0.1297	34.21% / 53.46% / 0.4172
7. HC-256 (2024 Panel 29 Disjoint)	**10,037 [1,258, 12.5%]**	**0% (DUPERSID & DUID)**	**0.7824 / 0.7812**	**0.3506**	**0.0951 / 0.0856**	**+0.1242**	**32.69% / 52.38% / 0.4026**
8. HC-257 (Panel 28 Prospective Year-2)	8,708 [1,157, 13.3%]	100% Panel 28	0.7824 / 0.7807	0.3543	0.1001 / 0.0874	+0.1263	34.72% / 53.33% / 0.4206

* Note: The HC-251 internal holdout set ($N = 3,755$) has 0% person-level (DUPERSID) overlap with Train, but 72.20% of holdout rows share a household (DUID) with training rows and the partition was reused across model/ablation evaluations. Row 3 (GroupKFold by DUID) evaluates zero-household-overlap CV (ROC-AUC = 0.7785), and Row 7 (HC-256 Panel 29, $N = 10,037$) provides strictly independent external validation with 0% person and 0% household overlap (at $\tau = 0.10$: Prec 23.78%, Rec 78.46%, F1 0.3650; at $\tau = 0.30$: Prec 42.29%, Rec 30.29%, F1 0.3529).

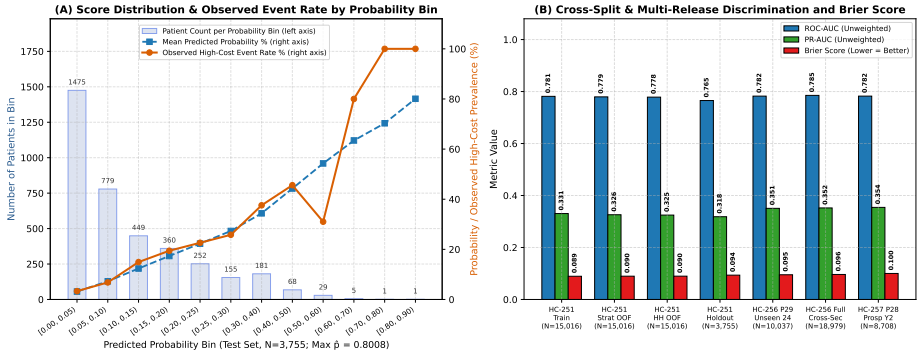


Fig. 11. (A) Internal holdout cohort probability score distribution and observed vs. predicted high-cost event rate by probability bin ($\max \hat{p}_{\text{test}} = 0.8008$) and (B) cross-split and multi-release discrimination and Brier score comparison across evaluated cohorts (showing stable discrimination across OOF and disjoint Panel 29, though not establishing the absence of overfitting).

- **High-Sensitivity Operating Point ($\tau = 0.10$ — Maximum Balanced Accuracy):** Maximizes Balanced Accuracy (0.7152 OOF, 0.7122 Holdout, 0.7121 Disjoint Panel 29) and captures 77.33% **of top-decile expenditure cases** (23.18% precision, 1.93× enrichment, flagging 39.97% unweighted / 33.54% weighted). Suitable when high sensitivity is prioritized and the per-case cost of false-positive flagging (1,153 FP) is low relative to missed high-cost cases (102 FN).
- **Primary-Grid F1-Optimal Operating Point ($\tau = 0.20$ — Balanced Precision–Recall Trade-Off):** Maximizes primary-grid F1 on OOF validation (OOF F1 = 0.3855 unweighted, 0.3551 weighted), achieving 30.49% **holdout precision** (2.54× **enrichment;** 32.69% **on independent Panel 29**), 46.89% **holdout recall** (50.23% **OOF,** 52.38% **Panel 29**), 85.45% **specificity**, and **F1** = 0.3695 (F1 = 0.4026 on Panel 29), while flagging

18.43% of the cohort (13.76% weighted) and **reducing false positives by** 58.28% ($1,153 \rightarrow 481$). An intermediate diagnostic cutoff of $\tau = 0.15$ yields higher sensitivity (62.44% recall, 26.71% precision, holdout F1 = 0.3742) when a 28.02% flagged volume is acceptable.

- **Higher-Specificity Operating Point ($\tau = 0.30$ — Reduced False-Positive Burden):** Achieves 40.00% **holdout precision (**3.34× **enrichment;** 42.29% **on independent Panel 29)**, 94.83% **specificity**, and 25.33% **recall (**30.29% **on Panel 29)** while flagging only the top 7.59% of individuals (5.69% weighted) and **cutting false positives by** 85.17% ($1,153 \rightarrow 171$), trading substantial recall for a lower false-positive rate under tight capacity constraints.

Explicit Distinction Between Decision Threshold ($\hat{p} \geq \tau$) and Ranked Cohort Cutoff. Evaluating $\hat{p}_i \geq \tau$ applies an individual probability cutoff (1,501 flagged at $\tau = 0.10$; 692 at $\tau = 0.20$; 285 at $\tau = 0.30$), whereas selecting the top 10% of individuals ranked by predicted risk (375 individuals) is a fixed-capacity decile cutoff (yielding $\approx 42\%$ precision and $\leq 35\%$ recall, close to operating between $\tau = 0.25$ [440 flagged] and $\tau = 0.30$ [285 flagged]).

6.6 Disclosed Model Weaknesses and Methodological Boundaries

1. **Dollar-Scale Prediction Limitations:** While tree-based and linear models fit on $\ln(1+y)$ achieve the lowest MAE on the median scale ($\approx \$7,516$), retransforming via $\exp(\hat{y})-1$ severely underpredicts the dollar mean by 72.8% (Mean Predicted = $\$2,384.13$ vs Observed = $\$8,764.62$), yielding negative dollar R^2 (-0.0134). The 11-feature Tweedie GLM ($p = 1.5$, log link) directly models un-logged dollars, achieving $R^2 = 0.0798$ (weighted $R^2 = 0.0832$) and tracking the aggregate mean ($\$8,177.92$ vs $\$8,764.62$); however, observed-to-predicted ratios vary across intermediate deciles (1.766× in Decile 1 to 0.789× in Decile 4).
2. **Dominance of the Econometric Persistence Benchmark:** In longitudinal prospective validation on MEPS HC-257 Panel 28 ($N = 8,708$), predicting Year 2 spending from Year 1 spending (`TOTEXPY1`) achieved an R^2 of 0.1204 (MAE = $\$8,625.76$), outperforming static diagnosis-only Tweedie GLM ($R^2 = 0.0652$). Self-reported diagnosis flags alone explain 6–8% of cost variance; past healthcare utilization remains the dominant predictor of future spending.
3. **Subgroup Discrimination Degradation:** On prospective validation (HC-257), high-cost classification discrimination drops from weighted ROC-AUC 0.7807 on the general population to 0.7466 on the adult SAQ subcohort with self-reported functional limitations, and supplementary fairness audits show poor discrimination in pediatric (< 18, ROC-AUC 0.40–0.49) and uniformly healthy cohorts.
4. **Absence of Decision-Curve Analysis and Standardized Clinical Indices:** Decision-curve analysis (DCA) was not performed, and public MEPS files omit granular 4- and 5-character ICD-10-CM codes required to compute Charlson or Elixhauser comorbidity indices.

Table 10. Parametric Value-at-Risk premium simulation sensitivity analysis ($N = 20,000$ draws per scenario, Moderate Risk tier $\hat{y} = \$5,000$).

Tier	Floor B	Loading α	Sim P5	Sim P50	Sim P95	Premium	Analytical Gamma P95	Error	Premium Uplift
Moderate ($5,000)	$0.00	0.00	$727.60	$4,007.41	**$13,116.94**	$13,024.55	+0.709%	Baseline ($0.00)	
Moderate ($5,000)	$0.00	0.10	$800.36	$4,408.15	**$14,428.63**	$14,327.00	+0.709%	+$1,311.69 (+10.00%)	
Moderate ($5,000)	$0.00	0.25	$909.50	$5,009.26	**$16,396.17**	$16,280.68	+0.709%	+$3,279.23 (+25.00%)	
Moderate ($5,000)	$0.00	0.50	$1,091.40	$6,011.11	**$19,675.41**	$19,536.82	+0.709%	+$6,558.47 (+50.00%)	
Moderate ($5,000)	$0.00	1.00	$1,455.20	$8,014.81	**$26,233.88**	$26,049.09	+0.709%	+$13,116.94 (+100.00%)	
Moderate ($5,000)	$500.00	0.00	$1,227.60	$4,507.41	**$13,616.94**	$13,524.55	+0.683%	Baseline ($0.00)	
Moderate ($5,000)	$500.00	0.10	$1,300.36	$4,908.15	**$14,928.63**	$14,827.00	+0.685%	+$1,311.69 (+9.63%)	
Moderate ($5,000)	$500.00	0.25	$1,409.50	$5,509.26	**$16,896.17**	$16,780.68	+0.688%	+$3,279.23 (+24.08%)	
Moderate ($5,000)	$500.00	0.50	$1,591.40	$6,511.11	**$20,175.41**	$20,036.82	+0.692%	+$6,558.47 (+48.16%)	
Moderate ($5,000)	$500.00	1.00	$1,955.20	$8,514.81	**$26,733.88**	$26,549.09	+0.696%	+$13,116.94 (+96.33%)	

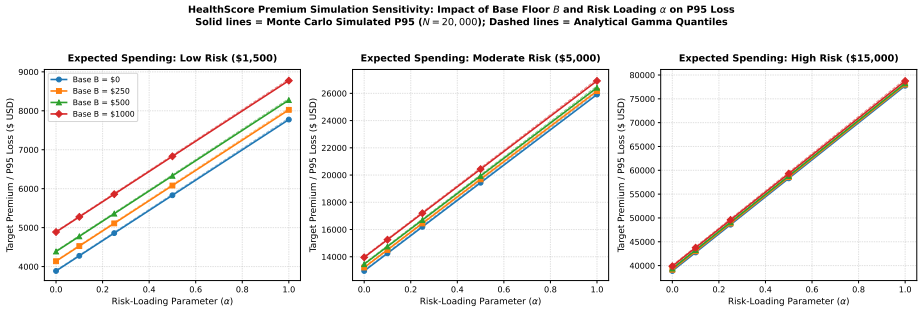


Fig. 12. HealthScore premium simulation sensitivity: Impact of base floor B and risk loading α on P95 loss across expected spending tiers. Solid lines represent Monte Carlo simulated P95 ($N = 20,000$); dashed lines represent analytical Gamma quantiles.

7 Configurable Loss and Premium Simulation (RQ5)

7.1 Mathematical Formulation and 60-Scenario Sensitivity Grid

In HealthScore, the output labeled `premium` is formalized as an exploratory parametric Value-at-Risk (VaR) simulation rather than an actuarially certified commercial insurance premium:

$$X_\alpha = B + (1+\alpha)G, \quad G \sim \text{Gamma}(k = 1.5, \theta = \hat{y}/1.5) \tag{12}$$

where Premium $= \text{Percentile}_{95}(X_\alpha)$, $B \geq 0$ is the configurable base-cost floor in USD (default: \$500.00), $E[G] = k\theta = \hat{y}$ is the Tweedie model-predicted expenditure, and $\alpha \geq 0$ is a dimensionless risk-loading parameter (default: $\alpha = 0.0$). For any quantile $q \in (0,1)$, the theoretical analytical quantile is $Q_{X_\alpha}(q) = B + (1+\alpha) \cdot \text{Gamma}^{-1}(q; k = 1.5, \theta = \hat{y}/1.5)$. We executed a grid sensitivity analysis across 4 base-cost floors ($B \in \{0, 250, 500, 1000\}$), 5 risk-loading levels ($\alpha \in \{0.0, 0.10, 0.25, 0.50, 1.00\}$), and 3 expenditure tiers ($\hat{y} \in \{\$1,500, \$5,000, \$15,000\}$) using Common Random Numbers ($N = 20,000$ draws per scenario; Table 10 and Fig. 12). Empirical Monte Carlo simulation tracks closed-form analytical Gamma quantiles within 0.68% **to** 0.71% across all 60 scenarios.

8 Integrated Discussion: Performance, Practical Use, and Trade-Offs

8.1 Consortium Throughput and Queueing Latency in Healthcare Operations

For interactive clinical workflows requiring sub-second response times (e.g., bedside vitals recording, prescription creation, or diagnostic scan orders), the consortium network should operate within the linear regime ($\leq$ 200 write TPS), where committed throughput tracks offered load linearly (199.9–200.2 TPS) with confirmation latency bounded between 0.03s and 0.06s. A major metropolitan health system processing 500,000 clinical encounters per day generates an average load of $\approx$ 6–10 TPS, peaking at $\approx$ 50–75 TPS—well within HealthScore's sub-100ms linear window. Under nominal 1,000 TPS stress load, the network sustains 350.1 TPS (18.00s latency) in the historical baseline, 515.8 TPS (6.98s latency) in the 5-worker rerun, and 485.27 ± 14.60 TPS (5.36 ± 0.22s latency) in the 8-worker repetitions with zero reported failures among submitted transactions. Because entering the saturation regime incurs multi-second queueing delays (5.2–18.0s), higher-throughput operation (470–515 write TPS) is appropriate only for asynchronous, delay-tolerant bulk ingestion protected by client-side rate-limiting queues. At 200 TPS continuous load ($\approx$ 1.5 KB/tx), append-only block storage expands by $\approx$ 1.08 GB/hour, whereas CouchDB world-state size remains bounded by active patient count $O(N_{\text{patients}})$.

8.2 Selecting Statistical Operating Points Under Resource Constraints

Our threshold sensitivity analysis clarifies the trade-offs involved in top-decile expenditure stratification. Operating at the high-sensitivity cutoff of $\tau = 0.10$ maximizes Balanced Accuracy (0.7122) by flagging 39.97% of the holdout cohort (1,501 individuals). The 29.98% theoretical volume ceiling represents the maximum achievable precision at this flagged cohort volume, while the model attains 23.18% precision and 77.33% recall. When downstream review capacity is constrained, shifting the decision boundary on the same calibrated probability output to the primary-grid F1-optimal operating point ($\tau = 0.20$) reduces false positives by 58.28% and raises holdout precision to 30.49% (32.69% on independent 2024 Panel 29) while retaining 46.89% recall (52.38% on Panel 29); shifting to the higher-specificity operating point ($\tau = 0.30$) reduces false positives by 85.17% and reaches 40.00% holdout precision (94.83% specificity) at the cost of capturing 25.33% of high-expenditure cases. Because maximizing F1 or adjusting τ does not establish clinical benefit or quantify the cost ratio of false positives to false negatives, actual deployment cutoffs should be calibrated to organization-specific resource budgets via formal decision-analytic evaluation.

9 Limitations and Threats to Validity

1. **Testbed-Specific Benchmark Environment, Absence of Fault-Injection Testing, and Queueing Latency Underestimation:** All Caliper benchmarks were conducted on a single host (macOS `arm64`) using Docker bridge networking and single administrative identity `HospitalOrgAdmin`. Multi-host WAN latency, cross-organization client concurrency, and physical fault-injection experiments (peer crashes, orderer partitions, disk corruption) were not evaluated. Furthermore, while the stochastic fluid-queueing model reproduces committed throughput within 0.41% at $1,000$ TPS, it underestimates historical stress-regime average latency by 48.5% (9.268s vs. 18.00s) and cannot be used as a validated predictor of saturation queue delay.
2. **Repeated Holdout Split Reuse, Unclustered Household Splitting, and Rotating Panel Overlap:** The MEPS HC-251 internal holdout partition ($N = 3,755$) was reused across the 17-scenario feature ablation and 9-model comparison evaluations rather than serving as an untouched final confirmatory test set, and row-level splitting in HC-251 shares $2,198$ household `DUID`s between Train and Holdout (72.20% of holdout rows). Furthermore, MEPS HC-256 ($N = 18,979$) shares rotating Panel 28 (46.24%) with HC-251, and HC-257 ($N = 8,708$) tracks Panel 28 longitudinally. We addressed these dependencies by evaluating 5-fold `GroupKFold(groups=DUID)` validation (ROC-AUC = 0.7785) and isolating the strictly disjoint 2024 HC-256 Panel 29 cohort ($N = 10,037$, zero person-level or household-level overlap, ROC-AUC = 0.7824) as our primary independent population benchmark; while these results show no evidence of a large generalization collapse across protocols, they do not prove the complete absence of overfitting.
3. **Non-Design-Based Bootstrap Uncertainty:** Reported 95% confidence intervals derive from non-parametric row-level bootstrap resampling ($B = 200$) and do not account for MEPS complex survey clustering (`VARPSU`) or stratification (`VARSTR`).
4. **Heuristic Intake Parsing, Absence of ICD-10 Comorbidity Indices, and Expenditure vs. Clinical Risk:** Runtime feature engineering relies on regex parsing of intake strings where unmentioned conditions default to zero, and public MEPS files omit granular 4- and 5-character ICD-10-CM codes required for Charlson or Elixhauser indices. Moreover, the classification target is top-decile annual expenditure ($\geq \$17,827.41$) rather than a validated clinical diagnosis or severity score.
5. **Direct Dollar Variance Ceiling, Aleatoric Acute Events, and Persistence Dominance:** Static diagnosis flags explain only $\approx 8\%$ of dollar expenditure variance ($R^2 = 0.0798$) and are outperformed by lagged Year-1 spending ($R^2 = 0.1204$). Furthermore, 10.22% of top-decile cases occur in individuals with zero baseline chronic conditions due to stochastic acute events, while 71.79% of individuals with ≥ 3 chronic clusters do not incur top-decile spending in a given year.

6. **On-Chain Identifiability, Absence of HIPAA/GDPR Certification, No Decision-Curve Analysis, and Exploratory VaR Status:** Although binary diagnostic reports and model weights remain off-chain, the prototype chaincode (`chaincode/index.js`) records plaintext `name`, `age`, `weight`, `height`, and unvalidated free-text strings (`history` in `createPatientRecord`, `remarks` in `createPrescription`, and `notes` in `requestScan`) in channel world state and relies on chaincode-level read redaction for `InsuranceOrgMSP` (returning `{id, age, healthScore, redacted: true}`); production deployments with real patient data would require structured codes, pseudonymous pointers, and off-chain encrypted storage. Neither HIPAA/GDPR compliance, comparative encrypted-database privacy benchmarks, nor decision-curve analysis (DCA) was assessed, and the Monte Carlo VaR simulation ($X_\alpha = B + (1+\alpha)G$) is strictly an exploratory technical demonstration uncertified for commercial underwriting or statutory solvency.

10 Conclusion

This paper presented an empirical performance, statistical, and systems evaluation of **HealthScore**, uniting a 5-organization permissioned Hyperledger Fabric blockchain consortium with an off-chain ACG-inspired multi-disease risk engine and an exploratory Value-at-Risk premium simulation. Our Caliper benchmarks demonstrate 199.9–200.2 write TPS with sub-100ms latency in the linear regime, 350.1–515.8 write TPS under stress saturation with zero reported transaction failures among submitted proposals, and 500.3–500.6 read TPS ($\leq$ 0.01s displayed average latency), while highlighting the $\approx$ 20–25 req/s single-worker off-chain ML inference boundary and disclosing that the fluid-queueing simulation underestimates saturation latency by 48.5%.

Across 46,458 MEPS records, controlled 17-scenario feature ablations confirm that custom ACG-inspired chronic-condition clusters provide the primary predictive lift over demographics (R^2 = 0.0395 $\rightarrow$ 0.0771, weighted BSS +0.0506 $\rightarrow$ +0.1025). Our deployed 11-feature Observable Logistic Regression classifier achieves ROC-AUC 0.7654, PR-AUC 0.3184, and weighted BSS +0.1025 on the repeatedly evaluated HC-251 internal holdout set ($N = 3,755$), and achieves **ROC-AUC** 0.7824 **(PR-AUC** 0.3506**, weighted BSS** +0.1242**) on our strongest independent population benchmark: the strictly disjoint 2024 MEPS HC-256 Panel 29 cohort (**$N = 10,037$**, with zero person-level or household-level overlap with HC-251)**. Through a 12-threshold sensitivity analysis ($\tau \in [0.05, 0.90]$) using 5-fold OOF cross-validation, we showed that at the reference high-sensitivity operating point of $\tau = 0.10$, the observed precision of 23.18% (77.33% recall) remains below the 29.98% theoretical upper bound imposed by the 39.97% flagged cohort volume (where 23.18%/29.98% = 77.33% equals model recall). Similar discrimination across stratified OOF (0.7795), household-grouped OOF (0.7785), and disjoint Panel 29 (0.7824) provides no evidence of a substantial generalization collapse under

the evaluated protocols, though it does not establish the absence of overfitting. Finally, characterizing statistical operating points at $\tau = 0.20$ (primary-grid F1-optimal: 30.49% holdout precision, 32.69% Panel 29 precision, 58.28% fewer false positives at 46.89% recall) and $\tau = 0.30$ (higher-specificity: 40.00% holdout precision, 94.83% specificity, 85.17% fewer false positives at 25.33% recall) alongside an analytically calibrated Value-at-Risk simulation provides a transparent, evidence-bounded foundation for permissioned healthcare workflow evaluation.

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